Summary- House Price Prediction

Key Drivers of Property Value: Based on the exploratory data analysis and our correlation heatmap, we identified clear primary indicators of a property's market price. The total square footage (area) is the most dominant factor, closely followed by the number of bathrooms and the presence of air conditioning. These specific elements consistently demonstrate a strong positive correlation with higher valuations.

Predictive Model Performance: The machine learning phase yielded solid baseline results. Our Linear Regression model achieved an R^2 score of approximately 0.65. In practical terms, this indicates that our model successfully accounts for 65% of the variance in real estate prices based purely on the selected features, providing a reliable mathematical foundation for automated price estimation.

Unexpected Analytical Findings: A notable surprise during the workflow was the pristine condition of the raw dataset, which contained zero missing values and lacked the severe outliers that typically skew real estate pricing data. Furthermore, the performance comparison between models was unexpectedly close; the straightforward, easily interpretable Linear Regression model remained highly competitive against the much more computationally complex Random Forest Regressor on our test set.

Strategic Business Recommendation: Translating these technical metrics into actionable business intelligence, my primary recommendation for a real estate firm is to heavily prioritize acquiring properties that boast large physical footprints and pre-installed air conditioning systems. If flipping or renovating a home, focusing budget allocation toward adding a bathroom or installing central air will likely yield the highest, most predictable return on investment during resale.